Alsu Bulatova

Frontend Engineer: Angular, TypeScript, React

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SUMMARY

Frontend engineer with four years of commercial experience, primarily in **Angular, TypeScript and RxJS**. I develop large existing applications, working with modular architecture, lazy loading, state management, design systems and change detection optimization. Recently migrated an application from Angular 18 to 22. I independently turn requirements into technical solutions, including API integration, tests and backend changes. English: B2.

EXPERIENCE

Liga Digital Economy, ***Frontend Engineer*** Dec 2025 - Present

Main project: enterprise AI platform*; Angular 18-22, TypeScript, RxJS, NGXS, HTML/SCSS, REST API, OpenAPI, WebSocket, Python/FastAPI, Vitest, Playwright, Karma/Jasmine, GitLab CI/CD, Docker*

Document search, AI chat, presentation and image generation, and visual agent configuration. The document-search chat serves 1,000+ enterprise users.

- **Migrated Angular 18 to 22 and upgraded to TypeScript 6**, resolving library incompatibilities and moving to standalone components with application- and route-level providers. Preserved authentication, lazy loading and user state restoration.
- Developed the **design system and shared UI components**: semantic colors, light and dark themes, inputs and styling controls. Standardized complex editors and implemented contextual onboarding.
- Built a **visual presentation editor** combining direct manipulation and AI commands. Managed document state and selection through NGXS, with undo and restoration of manual edits after reload.
- Implemented a **WebSocket editing protocol** with structured operations, acknowledgements and optimistic updates with rollback. Added server-side version checks and atomic operation batches, protecting newer edits from an earlier operation's rollback.
- Improved **authentication and guest access**: automatic token renewal, session expiry, sign-in recovery and guest session isolation. Built the UI for role-based tool permissions.
- Built **knowledge-base, index and document management** interfaces integrated with backend APIs. Set up typed client generation from OpenAPI and added server endpoints for inspecting document fragments.
- Optimized chat image loading with **IntersectionObserver** and language loading in highlight.js. Prevented stale asynchronous responses from replacing the currently selected document.
- Migrated tests from **Karma/Jasmine to Vitest**, retaining existing coverage and asynchronous scenarios. Migration validation covered **551 unit tests and 230 browser scenarios**, including authentication, editor gestures and error recovery.

Previous project: corporate travel platform*; TypeScript, FastAPI*

Flight, rail and hotel search and booking, fare and seat selection, and business-trip approvals.

- Owned frontend architecture and developed **search and booking flows**, including filtering, fare and seat selection, and protection against stale search results. Led a Feature-Sliced Design migration without pausing feature work.
- Extended **backend approval workflows** with atomic status transitions, bulk approvals with delegation, and cancellation rules protecting paid bookings. Addressed concurrent actions; critical flows were covered by Jest and Playwright.

EXPERIENCE - CONTINUED

Gazprombank ETP, ***Frontend Engineer*** Apr 2024 - Dec 2025

Cloud MDM platform for centralized product data, reference data and classifiers: large tables, validation, approval workflows and permissions. Angular, TypeScript, NGXS, RxJS, NG-Zorro, SignalR.

- Refactored a monolithic module with complex state and validation into a **lazy-loaded architecture**, reducing initial load time by **32%**.
- Migrated critical screens to **Angular Signals** and targeted reactive updates, improving rendering performance on large data tables by **15-20%**.
- Developed **role-based approval workflows** using NGXS state and RxJS data streams. Implemented real-time notifications through SignalR.
- Introduced reusable UI patterns and a standard for component data attributes, reducing QA automation rework by **60%**.
- Set up reproducible **Docker environments**, reducing environment setup blockers by **90%**. Mentored a backend engineer transitioning to Angular, reviewed code and participated in planning.

Avtomatika-Service (Gazprom Neft), ***Junior Frontend Engineer*** Oct 2022 - Apr 2024

Web applications for industrial process monitoring: equipment metrics, interactive installation diagrams and site configuration. Angular, TypeScript, RxJS, Apache ECharts, AntV X6.

- Built a **shared admin console for three applications** and contributed to the Module Federation rollout. Migrated disparate UI libraries to the PrizmUI design system.
- Developed **equipment monitoring dashboards and charts** with Apache ECharts, and visual schematic editors with custom nodes in AntV X6.
- Redesigned configuration workflows so analysts could handle part of the setup, reducing new site onboarding time by **40%**.
- Introduced **MSW API mocking** and Cypress component and E2E tests. Supported the React team with critical fixes, features and pre-release code reviews.
- Started an internal **frontend community** for knowledge sharing and consistent coding standards across teams. Created Figma prototypes to discuss and agree on interface solutions.

SKILLS AND ENGINEERING PRACTICES

Angular: Angular 18-22, TypeScript, RxJS, NGXS, Signals, standalone components, OnPush, change detection, lazy loading, modular architecture, Module Federation.
UI and APIs: HTML, CSS/SCSS, JavaScript, design systems, NG-Zorro, PrizmUI, REST APIs, OpenAPI, WebSocket. Also React, GraphQL, Python/FastAPI and SQL.
Quality and delivery: Vitest, Jest, Playwright, Cypress, MSW, Karma/Jasmine, code review, Git. Configured GitLab CI/CD and wrote Dockerfiles and service deployment configuration.
AI-assisted development: use Claude Code with explicit task scope, constraints and validation plans; review generated code and tests.

PERSONAL PROJECT

Koi - collaborative whiteboard*; React, TypeScript, Canvas, Go, GraphQL*

Built a custom rendering engine and multi-client synchronization with temporary object locks. Tested with Vitest and Playwright, including a two-client collaboration scenario.

INDEPENDENT LLM RESEARCH

I study **mechanistic interpretability** in Qwen models and agent behavior control through Inhibition Mask. This helps me understand model limitations in everyday development: what to delegate to a coding agent, where my oversight is needed and which failures to anticipate.

EDUCATION AND LANGUAGES

Gubkin Russian State University of Oil and Gas, BSc in Applied Mathematics, 2018-2022
Russian: native. English: B2.